Summer Project

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1 Background Information

1.1 Optical Physics

The optical beam is modelled as a Gaussian beam propagating along the optical axis (z-axis). Working in the paraxial approximation, the complex electric field is

$$E(x,y,z) = E_0 \frac{w_0}{w(z)} \exp\left(-\frac{x^2+y^2}{w(z)^2}\right) \exp\left(-ikz - i\frac{k(x^2+y^2)}{2R(z)} + i\zeta(z)\right), \tag{1}$$

where $k = 2\pi/\lambda$ is the wavenumber and the beam parameters are

$$w(z) = w_0\sqrt{1+\left(\frac{z}{z_R}\right)^2}, \tag{2}$$

$$R(z) = z\left(1+\left(\frac{z_R}{z}\right)^2\right), \tag{3}$$

$$\zeta(z) = \arctan\left(\frac{z}{z_R}\right), \qquad z_R = \frac{\pi w_0^2}{\lambda}. \tag{4}$$

Here w_0 is the beam waist, $w(z)$ is the beam radius at position z, $R(z)$ is the radius of curvature of the wavefronts, $\zeta(z)$ is the Gouy phase, and z_R is the Rayleigh range.

At the SLM plane $z = z_0$ the amplitude prefactor $w_0/w(z_0)$ is constant, and $\exp(-ikz_0)$ and $\exp(i\zeta(z_0))$ are *global* phases, so they may be dropped. The curvature term

$$\exp\left(-i\frac{k(x^2+y^2)}{2R(z_0)}\right)$$

is spatially varying and should be retained unless the beam is well collimated. The incident field at the SLM plane is therefore

$$E_{\text{in}}(x,y) = A\exp\left(-\frac{x^2+y^2}{w^2}\right)\exp\left(-i\frac{k(x^2+y^2)}{2R}\right), \tag{5}$$

where $w \equiv w(z_0)$, $R \equiv R(z_0)$, and A absorbs all constant prefactors. In the collimated limit ($R \to \infty$) this reduces to

$$E_{\text{in}}(x,y) = A\exp\left(-\frac{x^2+y^2}{w^2}\right). \tag{6}$$

The modulated field then propagates through thermo-optical material modelled by a linear transmission matrix T, and the detector measures only intensity:

$$E_{\text{out}} = T\,E_{\text{SLM}}, \qquad I(x,y) = |E_{\text{out}}(x,y)|^2. \tag{7}$$

Optical detectors measure intensity, not complex amplitude, so $I(x,y)$ is the only accessible quantity.

In summary, the incident field is the Gaussian beam evaluated at the SLM plane; spatially varying curvature phase is retained when needed; the downstream optical medium is captured by a complex transmission matrix; and the output camera measures intensity.

1.2 Thermal Physics

The optical medium is treated as a thermally perturbed phase/scattering element whose temperature field evolves on timescales much slower than the optical field. Over these slower thermal timescales, the temperature distribution $T(\mathbf{r}, t)$ is modelled by the heat equation

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (\kappa \nabla T) + Q(\mathbf{r}, t), \tag{8}$$

where ρ is the density, c_p is the specific heat capacity, κ is the thermal conductivity, and $Q(\mathbf{r}, t)$ is any applied or absorbed heat source.

Temperature affects the optical response through the thermo-optic effect. To first order, the refractive index is taken to vary linearly with temperature:

$$n(\mathbf{r}, t) = n_0 + \frac{dn}{dT}(T(\mathbf{r}, t) - T_0), \tag{9}$$

where n_0 is the reference refractive index at temperature T_0, and dn/dT is the thermo-optic coefficient of the material. These refractive-index variations change the optical path length through the medium and therefore introduce a time-dependent phase perturbation

$$\Delta\phi(x, y, t) = k \int \Delta n(x, y, z, t)\, dz, \tag{10}$$

with $k = 2\pi/\lambda$ and $\Delta n = n - n_0$.

At the system level, these thermally induced phase changes are represented as a slow drift in the effective optical operator of the medium. In the transmission-matrix description used here, this is written as

$$T_{\text{opt}}(t) = T_{\text{opt},0} + \Delta T_{\text{opt}}(t), \tag{11}$$

where $T_{\text{opt},0}$ is the nominal transmission matrix and $\Delta T_{\text{opt}}(t)$ captures the time-varying perturbation caused by thermal diffusion. The key thermal-physics problem is therefore to model how heat input produces a temperature field, how that temperature field modifies refractive index, and how the resulting refractive-index changes alter the optical response seen by the detector.

1.3 SLM Behaviour

The SLM acts only in the transverse (x, y) plane and is modeled as a phase-only modulator. The applied phase pattern $\phi(x, y)$ is the control input and sets the relative phases across the aperture, thereby altering interference at the output. The modulated field is

$$E_{\text{SLM}}(x, y) = E_{\text{in}}(x, y) e^{i\phi(x,y)}.$$

1.4 Physics-Informed Neural Networks

Physics-informed neural networks (PINNs) are a class of machine-learning models designed to approximate the solution of a physical system while explicitly respecting its governing equations. In a standard neural network, one typically provides input–output data pairs and trains the network to reproduce the observed mapping. A PINN extends this idea by requiring the network not only to fit data, but also to satisfy the differential equations, boundary conditions, and initial conditions that describe the underlying physics.

In the present project, the quantity of interest is the temperature field inside the thermo-optical material. Rather than solving for this field using only a conventional numerical discretisation, a PINN represents the temperature by a neural network

$$T_\theta(\mathbf{r}, t), \tag{12}$$

where $\mathbf{r}$ denotes spatial position, t denotes time, and θ represents the trainable parameters of the network, namely its weights and biases. The role of training is to adjust θ so that the network output approximates a temperature field that is both physically admissible and consistent with any available observations.

For the thermal model considered here, the governing equation is the heat equation

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (\kappa \nabla T) + Q(\mathbf{r}, t). \tag{13}$$

A PINN does not enforce this equation symbolically; instead, it enforces it approximately by penalising violations of the equation at many sampled points in space and time. Substituting the network output $T_\theta(\mathbf{r}, t)$ into the heat equation gives the residual

$$\mathcal{R}_\theta(\mathbf{r}, t) = \rho c_p \frac{\partial T_\theta}{\partial t} - \nabla \cdot (\kappa \nabla T_\theta) - Q(\mathbf{r}, t). \tag{14}$$

If T_θ were the exact physical solution, this residual would vanish everywhere. The PINN is therefore trained so that $\mathcal{R}_\theta(\mathbf{r}, t)$ is as small as possible over the domain of interest.

The overall training objective is written as a weighted sum of loss terms,

$$\mathcal{L} = \lambda_{\text{data}} \mathcal{L}_{\text{data}} + \lambda_{\text{PDE}} \mathcal{L}_{\text{PDE}} + \lambda_{\text{BC}} \mathcal{L}_{\text{BC}} + \lambda_{\text{IC}} \mathcal{L}_{\text{IC}}. \tag{15}$$

Each term plays a different role. The data loss $\mathcal{L}_{\text{data}}$ penalises disagreement between the network prediction and any available measurements. The PDE loss $\mathcal{L}_{\text{PDE}}$ penalises violations of the heat equation through the residual $\mathcal{R}_\theta$. The boundary-condition loss $\mathcal{L}_{\text{BC}}$ enforces the thermal conditions at the edges of the material, such as fixed temperature or insulated boundaries, and the initial-condition loss $\mathcal{L}_{\text{IC}}$ enforces the temperature field at the initial time.

The coefficients λ_{data}, λ_{PDE}, λ_{BC}, and λ_{IC} are scalar weighting factors that determine the relative importance of the different terms during training. They are not physical constants; rather, they are tunable hyperparameters chosen so that the network balances data fitting and physical consistency in a numerically stable way. For example, if λ_{PDE} is too small, the model may fit the data well but violate the heat equation significantly. If it is too large, the model may satisfy the equation while fitting observations poorly. In practice, these weights are selected empirically or adaptively to obtain a useful compromise between the competing objectives.

Training proceeds by sampling points in the space–time domain and evaluating the loss function at those points. The required derivatives, such as $\partial T_\theta / \partial t$ and spatial gradients of T_θ, are computed using automatic differentiation. An optimisation algorithm such as gradient descent or Adam then

updates the parameters θ to reduce the total loss. In this sense, the network "learns" by repeatedly proposing a candidate temperature field, measuring how strongly it violates the data and physics constraints, and adjusting its parameters to reduce those violations.

Once a thermally consistent approximation $T_\theta(\mathbf{r}, t)$ has been obtained, it can be coupled to the optical model through the thermo-optic relation

$$n_\theta(\mathbf{r}, t) = n_0 + \frac{dn}{dT}(T_\theta(\mathbf{r}, t) - T_0), \tag{16}$$

which gives a corresponding refractive-index field. This in turn induces a time-dependent optical phase perturbation,

$$\Delta\phi_\theta(x, y, t) = k \int (n_\theta(x, y, z, t) - n_0)\, dz, \tag{17}$$

and therefore modifies the effective optical response of the material. In the present setting, the PINN is thus used as a physics-constrained surrogate model for the thermal dynamics that drive slow drift in the optical system.

The main appeal of this approach is that it incorporates known thermal physics directly into the learning process. As a result, the model does not rely solely on large labelled datasets, and its predictions are encouraged to remain consistent with the governing equations of the system. This makes PINNs a natural candidate for modelling thermally driven changes in an optical medium whose internal state is only partially observed through optical measurements.

2 Experimental Setup

A setup of our system is shown below in Figure 1.

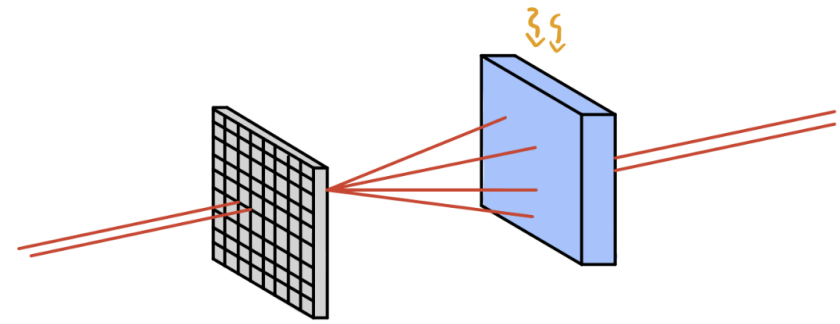

Figure 1: Schematic of the experimental setup. The laser beam is modulated by the SLM, passes through the some material, and is detected by the camera.

The optical process is summarized as follows:

i. Define the incident field $E_{\text{in}}(x, y)$ at the SLM plane.

ii. Apply SLM phase modulation to obtain $E_{\text{SLM}}(x, y) = E_{\text{in}}(x, y)e^{i\phi(x,y)}$.

iii. Propagate through the thermo-optical material using the transmission matrix $T_{\text{opt}}(t)$ to give $E_{\text{out}} = T_{\text{opt}}(t)E_{\text{SLM}}$.

iv. Compute the output intensity $I(x, y) = |E_{\text{out}}(x, y)|^2$.

v. Define the reinforcement-learning reward as the intensity at a selected target output location.